

DAPO Decoupled clipping & dynamic sampling for long-CoT RL

Sample a group



Math prompt + answer
q : 6 × 7 = ? a : 42



Rollout snapshot

$\pi_{\theta_{\text{old}}}$



6 responses / prompt
Fixed for the batch



Dynamic sampling

Retain groups with $0 < \text{accuracy} < 1$

1 0 1

Mixed → buffer Uniform outcomes → discard



Token-level clipped objective



Clip-Higher



Raise upper clip; retain lower clip

Clip the surrogate incentive.

Verify, shape & select



Completions



Reward

... 42 <eos> +1 ✓

... 17 <eos> -1 ✗

... 42 <eos> +1 ✓

Correctness + soft length reward

Group signal & trainable policy



Normalize rewards within each group

$$\hat{A}_{i,t} = \hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R)}$$

Share advantage across valid tokens



Current policy

π_{θ} Clipped policy update

No learned critic; no reference KL



Resample until the batch is full



Token-level reduction

Equal weight per valid token

+ + +
- - - - -

m = 0 for truncated responses



Soft overlong punishment

Correctness + length penalty

